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Exercise Sheet 4

This is not the official exercise sheet from the course.
More difficult exercises are denoted by ★ to ★★★

> If you can solve it, it is an exercise;
> otherwise it's a research problem.
>
> Richard Bellman

Exercise 1.
Give an example of a non-empty set A and a relation σ on A satisfying the following properties, or prove why it cannot exist.

1. Reflexive and irreflexive.
2. Symmetric and antisymmetric.
3. Symmetric, transitive, and reflexive.
4. Asymmetric and symmetric.
5. Reflexive, symmetric, and antisymmetric.
6. Irreflexive and symmetric.
7. Neither reflexive nor irreflexive.

Solution 1.

1. This is impossible. Since A is non-empty, choose $a \in A$. Reflexivity implies $(a, a) \in \sigma$, whereas irreflexivity implies $(a, a) \notin \sigma$, a contradiction.
2. Let $A = \{1, 2\}$ and let $\sigma = \{(1, 1)\}$. It is symmetric because its only related pair is a loop, and it is antisymmetric because it contains no pair (x, y) with $x \neq y$.
3. Let $A = \{1, 2\}$ and let $\sigma = A \times A$. Then σ is reflexive and symmetric. It is also transitive, since every ordered pair in $A \times A$ belongs to σ.
4. This is possible only for the empty relation. For an example, let $A = \{1\}$ and $\sigma = \varnothing$. The empty relation is symmetric and asymmetric, since both conditions are implications and there are no related pairs. Conversely, if a symmetric relation contained (a, b), symmetry would give (b, a), contradicting asymmetry. Thus any relation that is both must be empty.
5. Let $A = \{1, 2\}$ and $\sigma = \{(1, 1), (2, 2)\}$. Clearly σ is reflexive and symmetric. Antisymmetry holds because the only related pairs have equal entries. In fact, any relation with all three properties must be equality: symmetry and antisymmetry together imply that $a\sigma b$ is possible only when $a = b$, and reflexivity requires every (a, a) to belong to the relation.

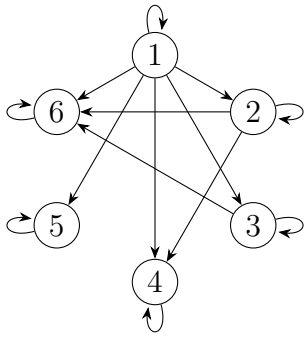
6. Let $A = \{1, 2\}$ and $\sigma = \{(1, 2), (2, 1)\}$. No pair of the form (a, a) belongs to σ, so it is irreflexive; the two arrows occur in opposite directions, so it is symmetric.
7. Let $A = \{1, 2\}$ and $\sigma = \{(1, 1)\}$. It is not irreflexive because $(1, 1) \in \sigma$, and it is not reflexive because $(2, 2) \notin \sigma$.

Exercise 2.
Let $A = \{1, 2, 3, 4, 5, 6\}$. Write out the relations corresponding to $|$, $\nmid$, $>$, and $\leq$, and express them in graph representation.

Solution 2.
For each relation, use the directed graph $G_R = (V, E_R)$ with vertex set $V = A = \{1, 2, 3, 4, 5, 6\}$ and an arrow $x \to y$ exactly when $(x, y) \in E_R$.

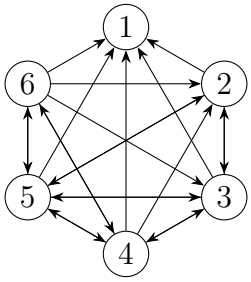
Divisibility Relation ($R_|$): The divisibility relation is

$$R_| = \{(1,1), (1,2), (1,3), (1,4), (1,5), (1,6), (2,2), (2,4), (2,6), (3,3), (3,6), (4,4), (5,5), (6,6)\}.$$



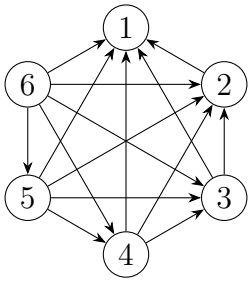
Non-Divisibility Relation ($R_\nmid$): The non-divisibility relation is the complement of $R_|$ in $A \times A$:

$$\begin{aligned} R_\nmid = \{&(2,1), (2,3), (2,5),\\ &(3,1), (3,2), (3,4), (3,5),\\ &(4,1), (4,2), (4,3), (4,5), (4,6),\\ &(5,1), (5,2), (5,3), (5,4), (5,6),\\ &(6,1), (6,2), (6,3), (6,4), (6,5)\}. \end{aligned}$$



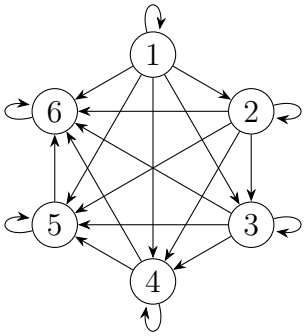
Greater-Than Relation ($R_>$): The greater-than relation is

$$\begin{aligned} R_> = \{&(2,1),\\ &(3,1),(3,2),\\ &(4,1),(4,2),(4,3),\\ &(5,1),(5,2),(5,3),(5,4),\\ &(6,1),(6,2),(6,3),(6,4),(6,5)\}. \end{aligned}$$



Less-Than-Or-Equal Relation ($R_\leq$): The less-than-or-equal relation is

$$\begin{aligned} R_\leq = \{&(1,1),(1,2),(1,3),(1,4),(1,5),(1,6),\\ &(2,2),(2,3),(2,4),(2,5),(2,6),\\ &(3,3),(3,4),(3,5),(3,6),\\ &(4,4),(4,5),(4,6),\\ &(5,5),(5,6),(6,6)\}. \end{aligned}$$



Exercise 3.
Prove or disprove: If σ and θ are relations on a set A, then

$$(\sigma \cup \theta)^* = \sigma^* \cup \theta^*.$$

Solution 3.
The statement is false in general. Let $A = \{1, 2, 3\}$ and choose

$$\sigma = \{(1, 2)\}, \qquad \theta = \{(2, 3)\}.$$

Since $(1, 2)$ and $(2, 3)$ both belong to $\sigma \cup \theta$, the reflexive-transitive closure $(\sigma \cup \theta)^*$ contains $(1, 3)$.

On the other hand, the only non-loop pair in σ^* is $(1, 2)$, and the only non-loop pair in θ^* is $(2, 3)$. Thus $(1, 3) \notin \sigma^* \cup \theta^*$. Consequently,

$$(\sigma \cup \theta)^* \neq \sigma^* \cup \theta^*.$$

Exercise 4.
Prove or disprove: If σ is a partial order on a set A, then its inverse relation σ^{-1} is also a partial order on A.

Solution 4.
The statement is true. Recall that

$$x \, \sigma^{-1} \, y \iff y \, \sigma \, x.$$

We check the three properties of a partial order.

Reflexivity. Since σ is reflexive, $x\sigma x$ for every $x \in A$. Hence $x\sigma^{-1}x$.

Antisymmetry. Suppose $x\sigma^{-1}y$ and $y\sigma^{-1}x$. Then $y\sigma x$ and $x\sigma y$. By antisymmetry of σ, $x = y$. Thus σ^{-1} is antisymmetric.

Transitivity. Suppose $x\sigma^{-1}y$ and $y\sigma^{-1}z$. Then $y\sigma x$ and $z\sigma y$. By transitivity of σ, $z\sigma x$, which means $x\sigma^{-1}z$. Thus σ^{-1} is transitive.

Therefore σ^{-1} is a partial order on A.

Exercise 5.
Let

$$P = \{\{0\}, \{-1, 1\}, \{-2, 2\}, \{-3, 3\}, \ldots\}$$

be a partition of $\mathbb{Z}$. Define an equivalence relation θ such that $\mathbb{Z}/\theta = P$.

Solution 5.
Define θ on $\mathbb{Z}$ by

$$x\theta y \iff |x| = |y|.$$

This is an equivalence relation: $|x| = |x|$ gives reflexivity; if $|x| = |y|$, then $|y| = |x|$, giving symmetry; and if $|x| = |y|$ and $|y| = |z|$, then $|x| = |z|$, giving transitivity.

For each $n \in \mathbb{Z}$,

$$[n]_\theta = \{y \in \mathbb{Z} : |y| = |n|\} = \begin{cases} \{0\}, & n = 0, \\ \{-|n|, |n|\}, & n \neq 0. \end{cases}$$

Thus the equivalence classes are exactly $\{0\}, \{-1, 1\}, \{-2, 2\}, \ldots$, and so $\mathbb{Z}/\theta = P$.

Exercise 6.
Let σ be a relation on $\mathbb{Z}$ defined by

$$x\sigma y \iff x^3 + y^3 \text{ is even.}$$

Prove that σ is an equivalence relation and describe its equivalence classes.

Solution 6.
For every integer x, we have $x^3 \equiv x \pmod 2$. Therefore

$$x\sigma y \iff x^3 + y^3 \equiv 0 \pmod 2 \iff x + y \equiv 0 \pmod 2.$$

The last condition says precisely that x and y have the same parity.

Reflexivity. Every integer has the same parity as itself, so $x\sigma x$.

Symmetry. If x and y have the same parity, then y and x have the same parity. Equivalently, $x^3 + y^3 = y^3 + x^3$ is even whenever it is even.

Transitivity. If $x\sigma y$ and $y\sigma z$, then x and y have the same parity, and y and z have the same parity. Hence x and z have the same parity, so $x\sigma z$.

Thus σ is an equivalence relation. There are exactly two equivalence classes:

$$[0]_\sigma = 2\mathbb{Z}, \qquad [1]_\sigma = 2\mathbb{Z} + 1,$$

the even integers and the odd integers, respectively.

Exercise 7.
Let σ be a relation on $\mathbb{R} \setminus \{0\}$ defined by

$$x\sigma y \iff \frac{x}{y} \in \mathbb{Q}^+.$$

Prove that σ is an equivalence relation and describe its equivalence classes. Are $\sqrt{2}$ and $\sqrt{8}$ in the same equivalence class? What about $\sqrt{2}$ and $\sqrt{3}$?

Solution 7.
The domain excludes zero, so all ratios below are well-defined.

Reflexivity. For every $x \neq 0$, $x/x = 1 \in \mathbb{Q}^+$, so $x\sigma x$.

Symmetry. If $x\sigma y$, then $x/y = q$ for some $q \in \mathbb{Q}^+$. Thus $y/x = 1/q \in \mathbb{Q}^+$, so $y\sigma x$.

Transitivity. If $x\sigma y$ and $y\sigma z$, then x/y and y/z are positive rational numbers. Therefore

$$\frac{x}{z} = \frac{x}{y}\frac{y}{z} \in \mathbb{Q}^+,$$

so $x\sigma z$. Hence σ is an equivalence relation.

For any $x \neq 0$, its equivalence class is

$$[x]_\sigma = \{y \in \mathbb{R} \setminus \{0\} : x/y \in \mathbb{Q}^+\}$$

Since $\sqrt{2}/\sqrt{8} = 1/2 \in \mathbb{Q}^+$, the numbers $\sqrt{2}$ and $\sqrt{8}$ are equivalent. But

$$\frac{\sqrt{2}}{\sqrt{3}} = \sqrt{\frac{2}{3}} \notin \mathbb{Q},$$

so $\sqrt{2}$ and $\sqrt{3}$ are not equivalent. (Indeed, if $\sqrt{2/3}$ were rational, then $3\sqrt{2/3} = \sqrt{6}$ would be rational, a contradiction.)

Exercise 8.
Let $A = \mathbb{N} \times \mathbb{N}$. Define the lexicographical relation $\preceq$ on A by

$$(a, b) \preceq (c, d) \iff (a < c) \text{ or } (a = c \text{ and } b \leq d).$$

Prove that $\preceq$ is a total order on A.

Solution 8.
We prove reflexivity, antisymmetry, transitivity, and totality.

Reflexivity. For every $(a, b) \in A$, we have $a = a$ and $b \leq b$. Hence $(a, b) \preceq (a, b)$.

Antisymmetry. Suppose $(a, b) \preceq (c, d)$ and $(c, d) \preceq (a, b)$. The first relation implies $a \leq c$, and the second implies $c \leq a$, so $a = c$. The defining condition now gives $b \leq d$ and $d \leq b$, hence $b = d$. Therefore $(a, b) = (c, d)$.

Transitivity. Suppose $(a, b) \preceq (c, d)$ and $(c, d) \preceq (e, f)$. Then $a \leq c \leq e$, so $a \leq e$. If $a < e$, then $(a, b) \preceq (e, f)$ by definition. The remaining possibility is $a = e$. In that case, $a \leq c \leq e$ forces $a = c = e$. The two assumed relations then imply $b \leq d$ and $d \leq f$, so $b \leq f$. Thus $(a, b) \preceq (e, f)$ in this case as well.

Totality. Take any $(a, b), (c, d) \in A$. Since the usual order on $\mathbb{N}$ is total, either $a < c$, $c < a$, or $a = c$. In the first case $(a, b) \preceq (c, d)$; in the second case $(c, d) \preceq (a, b)$. If $a = c$, then either $b \leq d$ or $d \leq b$, so at least one of $(a, b) \preceq (c, d)$ and $(c, d) \preceq (a, b)$ holds.

Thus $\preceq$ is a total order on A.

★ **Exercise 9.**
Prove or disprove: Every partition of a set A is equal to A/θ for some equivalence relation θ on A.

Solution 9.
The statement is true. Let P be any partition of A. Define a relation θ on A by

$$x \theta y \iff x \text{ and } y \text{ belong to the same block of } P.$$

This is reflexive because every element belongs to a block of P, and so it belongs to the same block as itself. It is symmetric because if x and y belong to the same block, then y and x do too. It is transitive because if x, y belong to one block and y, z belong to one block, those two blocks must be the same: distinct blocks of a partition are disjoint, and both contain y. Thus x, z belong to the same block.

Hence θ is an equivalence relation. For every $x \in A$, its equivalence class $[x]_\theta$ is exactly the unique block of P that contains x. Therefore the collection of all equivalence classes is precisely P, and $A/\theta = P$.

★ **Exercise 10.**
Let σ be any relation on a non-empty set A. Let $\theta = \sigma \cup \sigma^{-1} \cup \mathrm{Id}_A$.

1. Prove that θ^* is an equivalence relation.
2. Prove that θ^* is the smallest equivalence relation containing σ: if ρ is any equivalence relation on A such that $\sigma \subseteq \rho$, then $\theta^* \subseteq \rho$.

Solution 10.
Recall that the transitive closure of θ is defined as

$$\theta^* = \bigcup_{n \in \mathbb{N} \setminus \{0\}} \theta^n,$$

where $\theta^1 = \theta$ and $\theta^{n+1} = \theta^n \circ \theta$.

1. **Equivalence relation:**
   - *Transitivity:* By definition, the transitive closure θ^* is transitive because $\theta^* \circ \theta^* \subseteq \theta^*$.
   - *Symmetry:* Since $\theta = \sigma \cup \sigma^{-1}$, we have $\theta^{-1} = (\sigma \cup \sigma^{-1})^{-1} = \sigma^{-1} \cup \sigma = \theta$, meaning θ is symmetric. By induction, every power θ^n is symmetric ($\theta^n = (\theta^n)^{-1}$), so their union $\theta^* = \bigcup_{n \in \mathbb{N} \setminus \{0\}} \theta^n$ is also symmetric.
   - *Reflexivity:* θ^* is reflexive (incorporating reflexivity through the relation structure). It is therefore an equivalence relation.
2. **Smallest equivalence relation containing σ:**

   Since $\sigma \subseteq \theta = \theta^1 \subseteq \theta^* = \bigcup_{n \in \mathbb{N} \setminus \{0\}} \theta^n$, the relation θ^* contains σ. Now let ρ be any equivalence relation on A such that $\sigma \subseteq \rho$. Because ρ is symmetric, $\sigma^{-1} \subseteq \rho^{-1} = \rho$, which implies that

   $$\theta = \sigma \cup \sigma^{-1} \subseteq \rho.$$

   Furthermore, since ρ is transitive, by induction $\theta^n \subseteq \rho$ for all $n \in \mathbb{N} \setminus \{0\}$. Taking the union over all n yields

   $$\theta^* = \bigcup_{n \in \mathbb{N} \setminus \{0\}} \theta^n \subseteq \rho.$$

   Thus, θ^* is the smallest equivalence relation containing σ.